

Preliminary Research Document – Group 8

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Provider Cost Report Data (2011–2022)

Document Type: Preliminary Research Summary

Location: St. Clair College

Section 1: Introduction

The financial reporting structure of U.S. hospitals is fragmented, with most publicly available datasets offering state-level or system-level summaries. As a result, critical insights at the individual hospital level remain unclear.

The CMS Hospital Provider Cost Report dataset helps address this gap by offering hospital-level details, including:

- Costs and charges
- Patient volume
- Medicare/Medicaid utilization
- Operational measures and demographics.

These details enable comparative analysis across hospital size, ownership type, rural/urban classification, and geographic location, helping uncover meaningful patterns in hospital performance.

Section 2: Research Problem

U.S. hospitals must balance financial sustainability while providing equitable patient care. However, aggregated data makes it difficult to identify specific operational or financial challenges.

Examples of hidden problem areas include:

- Rural hospitals often display weaker financial stability than urban hospitals.
- High dependence on Medicare/Medicaid can influence long-term viability.
- Cost-to-charge ratios may reveal inefficiencies or under-reimbursement issues.

The lack of granular hospital-level insights limits data-driven decision-making for policymakers, administrators, and healthcare leaders. This research aims to address these visibility gaps.

Section 3: Research Objectives

The objectives of this project are to:

- Analyze hospital financial and operational performance using descriptive and predictive methods.
- Identify trends in cost-to-charge ratios and evaluate differences between hospital categories.
- Compare hospitals across factors such as location (rural vs. urban), ownership type, and size.
- Understand how reliance on Medicare and Medicaid affects hospital financial outcomes.
- Explore predictive modeling approaches such as regression, clustering, and time-based forecasting.

Section 4: Research Questions

Key research questions guiding this study include:

1. How do hospital financial outcomes differ by type, size, and geographic location?
2. How does Medicare/Medicaid dependence influence hospital performance?
3. What trends exist in cost-to-charge ratios over the 2011–2022 period?
4. Do rural hospitals face distinct financial challenges compared to urban hospitals?
5. Can predictive models help forecast financial or operational trends for hospitals?

Section 5: Methodology and Framework

Data Collection: Use the CMS-published Hospital Provider Cost Report dataset (117+ variables).

Data Cleaning:

- Handle missing values and inconsistencies.
- Standardize formats (numeric, categorical, date fields).
- Validate data quality and remove anomalies when necessary.

Exploratory Data Analysis (EDA):

- Summary statistics for costs, revenues, expenses, and utilization.
- Variance analysis by hospital type, size, ownership, and location.
- Trend identification for Medicaid/Medicare reliance.

Modeling Approaches:

- Regression modeling (financial predictors)
- Clustering (hospital grouping patterns)
- Forecasting (trend prediction)
- Geographic mapping (spatial insights)

Visualization Tools:

- Power BI dashboards (financial, operational, geographical, and combined overview)

Section 6: Initial Phase Findings

In the initial phase of analysis, the team will:

- Evaluate dataset completeness and accuracy.
- Generate basic statistics for core financial and operational variables.
- Compare performance between rural and urban hospitals.
- Identify emerging patterns related to Medicare/Medicaid reliance.
- Build preliminary regression and clustering models.
- Document early findings to guide the next stages of analysis.